WEEK 2 PROJECT

DASHBOARD VISUALIZATION

Build an interactive business dashboard using cleaned retail sales data

Project	Superstore Sales & Profit Dashboard
Recommended tool	Microsoft Power BI Desktop (free desktop app)
Input data	Cleaned Superstore CSV from the EDA / data-cleaning task
Dashboard features	KPI cards, 5 charts, slicers and written insights
Final hand-in	Power BI file, PDF/screenshot, cleaned data and README

Beginner-friendly step-by-step guide. The screenshots and numeric findings must come from your own Power BI report after loading the dataset.

1. Project brief

Objective: Build a simple interactive dashboard in Power BI using the cleaned Superstore sales dataset. The dashboard should summarise business performance and allow a user to explore sales and profit by date, product category, region and customer segment.

One-sentence hand-in: I will submit an interactive Power BI dashboard built from cleaned sales data, including KPI cards, charts, filters, key business insights and a short README.

What is a dashboard? A dashboard is a visual screen that combines important measures and charts in one place so users can monitor performance and make decisions quickly.

2. Recommended tool and input data

Recommended: Microsoft Power BI Desktop. It is suitable for a beginner, supports CSV files, offers interactive visuals and can save the project as a `.pbix` file.

Download Power BI Desktop from the official Microsoft Power BI Desktop page. If you use Tableau or Looker Studio instead, see Section 11.

Input: Use the cleaned Superstore CSV created in your previous EDA/data-cleaning task. If you used the guide from the previous task, the filename may be `StudentsPerformance_Cleaned.csv` for the student dataset or `SampleSuperstore.csv` for the retail dataset. **This dashboard project guide is designed for the Superstore retail dataset.** Use the cleaned Superstore file (for example, `SampleSuperstore_Cleaned.csv` or the original Superstore CSV after your cleaning steps). Do not use the student-score dataset for this sales dashboard.

If you have not created a cleaned Superstore file yet, download the Sample Superstore dataset from Kaggle – Sample Superstore, clean it in your previous task, and save the result as **SampleSuperstore_Cleaned.csv**.

3. Fields needed in the dashboard

Field	Purpose
Order Date	Time trend and date filtering
Sales	Revenue KPI and sales charts
Profit	Profit KPI and profit charts
Quantity	Units sold KPI
Category / Sub-Category	Product performance
Region / State	Geographical performance
Segment	Customer group comparison
Discount	Discount analysis

4. Dashboard layout

Create one report page with a clear title at the top, KPI cards directly underneath, charts in the middle, and slicers along the top or left side. Keep the layout uncluttered and use consistent colours.

Area	Visual	What it displays
Top title	Text box	Superstore Sales & Profit Dashboard
KPI row	4 Card visuals	Total Sales, Total Profit, Orders, Units Sold
Main left	Line chart	Sales by month
Main right	Clustered bar chart	Sales by category

Area	Visual	What it displays
Bottom left	Bar chart	Profit by region
Bottom middle	Bar chart	Profit by sub-category
Bottom right	Donut chart	Sales by customer segment
Filter area	Slicers	Year, Region, Category, Segment

5. Step-by-step: build it in Power BI

Step 1 – Open Power BI Desktop. Start Power BI Desktop. Select **Home** → **Get data** → **Text/CSV**. Browse to SampleSuperstore_Cleaned.csv. Select the file, check the preview and click **Load**.

Step 2 – Check the data. Select **Data view** and inspect the columns. Confirm Sales, Profit, Quantity and Discount are numeric. Set Order Date to the Date data type. If types are wrong, select the column and change Data type from the ribbon.

Step 3 – Create measures. In the Fields/Data pane, right-click the table and choose **New measure**. Create the DAX measures in Section 6 one by one. Use your actual table name in place of 'SampleSuperstore' if Power BI named it differently.

Step 4 – Add KPI cards. In Report view, select the Card visual. Add Total Sales, Total Profit, Orders and Units Sold as four separate cards. Format Sales and Profit as currency or a suitable number format.

Step 5 – Add charts. Add the five visuals listed in Section 7. Drag the correct fields into each visual's Axis, Values, Legend or X/Y field wells. Give each chart a meaningful title.

Step 6 – Add slicers. Add slicer visuals for Year, Region, Category and Segment. For Year, use the Order Date hierarchy or create a Year column using the DAX in Section 6. Test each slicer to ensure it filters the charts.

Step 7 – Format the page. Align the visuals, use readable labels and consistent colours. Add a clear dashboard title. Avoid too many colours, tiny text or unnecessary decoration.

Step 8 – Save and export. Save the report as Superstore_Sales_Dashboard.pbix. Use File → Export → Export to PDF if available, or take a clear screenshot of the report page. Keep the PBIX file as your editable project.

6. DAX measures (copy one at a time)

DAX is the formula language used in Power BI to calculate measures and columns. These measures assume your imported table is named **SampleSuperstore**. If the name is different, replace it with the exact table name shown in the Fields pane.

```
Total Sales = SUM('SampleSuperstore'[Sales])

Total Profit = SUM('SampleSuperstore'[Profit])

Units Sold = SUM('SampleSuperstore'[Quantity])

Orders = DISTINCTCOUNT('SampleSuperstore'[Order ID])

Profit Margin % =
DIVIDE([Total Profit], [Total Sales], 0)

Order Year = YEAR('SampleSuperstore'[Order Date])
```

Important: The first five are measures. Create `Order Year` as a calculated column using **New column**, not New measure. If your file has no Order ID field, use `COUNTROWS('SampleSuperstore')` as a simple order-line count and label it clearly as “Order Lines”, not “Orders”.

7. Required visuals and field setup

Visual	Fields to drag	Purpose
Card – Total Sales	[Total Sales]	Total revenue
Card – Total Profit	[Total Profit]	Overall profit
Card – Orders	[Orders]	Distinct order count

Visual	Fields to drag	Purpose
Card – Units Sold	[Units Sold]	Total quantity
Line chart	X-axis: Order Date (Month); Y-axis: [Total Sales]	Monthly sales trend
Clustered bar	Axis: Category; Values: [Total Sales]	Compare category sales
Bar chart	Axis: Region; Values: [Total Profit]	Compare regional profit
Bar chart	Axis: Sub-Category; Values: [Total Profit]	Find profitable / loss-making products
Donut chart	Legend: Segment; Values: [Total Sales]	Compare customer segments

8. Add interactive filters (slicers)

• **Year:** use Order Year or the Year level of the Order Date hierarchy.

• **Region:** use Region.

• **Category:** use Category.

• **Segment:** use Segment.

To add a slicer: select a blank area on the report canvas, choose the Slicer visual, and drag the required field into it. Select one or more values and check that all relevant charts and cards update. Use Format → Edit interactions if a slicer does not affect a visual.

9. Business insights to write

Write 4–6 insights after interacting with your dashboard. Use your actual dashboard results; do not invent figures. These sentence starters will help:

• “The category with the highest sales is _____, with total sales of _____.”

• “The most profitable region is _____, with total profit of _____.”

• “The sub-category _____ has the lowest profit / a loss, so its pricing and discount strategy should be reviewed.”

• “Sales increased/decreased during _____ compared with _____.”

• “The _____ customer segment contributes the largest share of total sales.”

• “When the Region slicer is changed to _____, the dashboard shows _____.”

A good business insight states what the chart shows and why it matters. For example: “The Technology category contributes the highest revenue, so the business should examine whether it also has a strong profit margin before increasing its marketing budget.” Only use this example if your actual dashboard supports it.

10. Testing checklist

■ The cleaned Superstore CSV loads successfully.

■ Sales, Profit and Quantity are numeric; Order Date is a date.

■ All four KPI cards show values and respond to filters.

■ All five charts display correct labels and titles.

■ Year, Region, Category and Segment slicers filter the report correctly.

■ No visual shows an unexplained blank or error.

■ The dashboard is readable and not overcrowded.

■ The PBIX file is saved and the PDF or screenshot is exported.

■ Business insights match the actual dashboard values.

11. If you use Tableau or Looker Studio

Tableau: Open Tableau Public/Desktop → Connect → Text file → select the cleaned CSV. Drag Order Date to Columns and Sales to Rows for a line chart. Create additional sheets for category sales, regional profit, sub-category profit and segment sales. Add KPI sheets and filters for Year, Region, Category and Segment. Combine the sheets in **New Dashboard**, then save or publish your work.

Looker Studio: Open Looker Studio → Create → Report → Add data. Connect a Google Sheets copy of the cleaned CSV or another supported source. Add scorecards for Sales, Profit, Orders and Quantity; add time series and bar/donut charts; then add date-range, Region, Category and Segment controls. Share or download the report as PDF if your account permits.

Use only one tool for this assignment unless your mentor specifically asks for more. Power BI is the recommended route in this guide.

12. README.md – copy into your project

```
# Week 2: Dashboard Visualization

## Project
Superstore Sales & Profit Dashboard built using Microsoft Power BI.

## Objective
Create an interactive dashboard from cleaned retail sales data to monitor sales, profit, orders, quantity and product/customer/region perf

## Tool
Microsoft Power BI Desktop.

## Input
Cleaned Superstore CSV file (for example, `SampleSuperstore_Cleaned.csv`).

## Dashboard components
- KPI cards: Total Sales, Total Profit, Orders, Units Sold
- Charts: monthly sales, sales by category, profit by region,
  profit by sub-category, sales by customer segment
- Slicers: Year, Region, Category, Segment

## Files to submit
- `Superstore_Sales_Dashboard.pbix`
- Dashboard PDF or screenshot
- Cleaned CSV dataset
- This README

## Learning
I learned how to import cleaned data, create DAX measures, build visualisations,
add slicers, test interactions and communicate business insights through a dashboard.
```

13. Final submission note

> “I completed the Week 2 Dashboard Visualization project using Microsoft Power BI and the cleaned Superstore sales dataset. I created KPI cards to display total sales, total profit, orders and units sold. I added charts to analyse monthly sales, product categories, regional profit, sub-category performance and customer segments. I also added interactive slicers for year, region, category and segment. This project helped me learn data visualisation, basic DAX measures, dashboard design and business insight communication.”

Submit these files: Superstore_Sales_Dashboard.pbix, dashboard PDF or screenshot, cleaned Superstore CSV, README.md, and this project guide PDF. If your portal accepts only one attachment, create a ZIP of the project files if allowed.

14. Learning resources

- Microsoft Learn – Power BI
- Power BI Desktop download
- Looker Studio
- Tableau learning

End of Week 2 Dashboard Visualization project guide.